Offline rehearsal guide

Portaldot Hacker House 2026
Node Operations & Tooling

What you can verify today

PortalPulse separates network reachability, identity, transaction inclusion, execution, finality and caller-specific state readback. Six deterministic scenarios let you reproduce the failure boundaries without a wallet or testnet.

Available now	Evidence scope
Readonly console	Explicit simulation, historical report import and manual readonly profiles. No default live endpoint.
Execution rehearsal	Six named synthetic inputs evaluated by the real strict checker. No canned success labels.
Portable reports	Versioned JSON, original facts/times, exact integers, SHA-256 and strict reimport.
Current progress submission	Build50050 submitted on 20 October; last actual platform readback: Under Review.

Read the boundaries first

Normal means the supplied facts agree internally. It does not prove source authenticity, independent consensus, a trusted network identity or a successful real transaction. Every rehearsal remains simulated and offline:

```
origin=simulated       mode=offline
verifiedNetwork=false  writeAllowed=false
fullVerification=false
```

No recognized-V3 probe deployment or complete live execution is claimed. Submission is not organizer approval or final delivery. This guide is project documentation, not a platform-issued certificate.

Open the project

Public prototype / Materials index / DoraHacks Build50050

github.com/JokerFuFu/portaldot-hacker-house-2026-portalpulse

Try six outcomes

In the webpage, switch to execution review. The rehearsal selector automatically loads the chosen case. The load button reloads your current selection.

Scenario ID	Expected result	Exit
complete	All six supplied stages agree internally. Verification/write flags remain false.	0
missing-evidence	Execution is absent. Later supplied facts cannot replace it by inference.	2
execution-failed	The transaction is included, but its explicit execution failure stays abnormal.	1
wrong-transaction	A success result belonging to a different transaction cannot pass correlation.	1
unfinalized	Execution succeeded; finality and state-readback evidence are absent. Still unverified.	2
state-mismatch	Expected and observed sequences differ by1; retain both exact integers.	1

A 90-second browser flow

1. Load complete. Inspect the six stages and the explicit simulated/offline source.
2. Select execution-failed and wrong-transaction. Both must be abnormal for different reasons.
3. Select missing-evidence and unfinalized. Missing facts remain unverified.
4. Select state-mismatch. Compare expected 9007199254740993 with observed 9007199254740992.
5. Export the execution report and reimport that JSON. Facts, time and digest remain intact.
6. Alter a payload field without updating its digest. Reimport must reject it and retain the last valid report.

Synthetic observation time remains 2026-10-14T00:00:00.000Z. Rehearsal does not relabel old facts as current network observations. The editor and controls do not print with the report.

Reproduce and review

Run the same checker locally

Use Node.js >=22.12 and the repository lockfile. No wallet, RPC config or account is required for these rehearsal commands.

```
npm --userconfig /dev/null --registry https://registry.npmjs.org \
  ci --ignore-scripts
npm test
npm run typecheck
npm run build
node dist/cli/main.js execution-demo --scenario complete
node dist/cli/main.js execution-demo --scenario state-mismatch \
  --out /private/tmp/pp-mismatch.json
node dist/cli/main.js execution-inspect \
  --input /private/tmp/pp-mismatch.json
```

Complete exits 0; missing evidence/unfinalized exit 2; abnormal evidence exits 1 with a diagnostic JSON summary. Abnormal and incomplete reports are still saved. Use a new output path on repeats: existing files are never overwritten. Malformed options/input or output errors emit stderr and no successful summary.

What was checked for this increment

Behavior revision 5ad3fe6 passed 99 focused and 507 full Node22.12 tests, both type checks and the web build. The actual local browser rendered all six expected outcomes and retained offline/three-false flags. Dependencies were reused; no new clean-install claim is made.

October16 A4 samples, eight-slide deck and 92-second captioned snapshot video keep their original dates and synthetic scope. They do not claim this selector or live execution. The October17 real monitoring report remains a separate historical RPC record.

What must happen before live proof

Confirm independently trusted organizer-recognized V3 Test identity against live block0. Then verify the selected wallet origin/mapping/funds, compiler target and bounded gas/storage exposure. Only after those checks does the user sign. Retain the deployment/update receipts, exact execution, linked finality and anchored state readback.

The final integration video and final DoraHacks submission remain outstanding. This rehearsal does not send a message, sign or broadcast a transaction.

Sources and exact scope

Reviewer instructions and six-case matrix
Dated development and verification evidence
Official event requirements